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MySQL Stored Procedures

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What is a Stored Procedure?

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A stored procedure is a precompiled SQL code that can be saved and reused.

If you have an SQL query that you write over and over again, save it as a stored procedure, and then just call it to execute it.

A stored procedure can also have parameters, so it can act based on the parameter value(s) that is passed.

Key Benefits of Stored Procedures

Stored procedures are widely used in database management, and have the following benefits:

- Code Reusability - The same procedure can be called from various applications
- Improved Performance - Stored procedures are precompiled and runs faster

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Stored Procedure Syntax

```
DELIMITER //

CREATE PROCEDURE procedure_name
  @param1 datatype,
  @param2 datatype
BEGIN
  -- SQL_statements to be executed
  SELECT column1, column2
  FROM table_name
  WHERE columnN = @paramN;
END //

DELIMITER;
```

Execute a Stored Procedure

To run a stored procedure, use the `CALL` statement:

```
CALL procedure_name('value1', 'value2');
```

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Drop a Stored Procedure

To delete a stored procedure, use the `DROP PROCEDURE` statement:

```
DROP PROCEDURE procedure_name;
```

Tip: To ensure that `DROP PROCEDURE` does not return an error, if the procedure is missing, add the `IF EXISTS` clause:

```
DROP PROCEDURE IF EXISTS procedure_name;
```

Demo Database

In this tutorial we will use the well-known Northwind sample database.

Below is a selection from the "Customers" table in the Northwind sample database:

CustomerID	CustomerName	ContactName	Address	City	PostalCode	Co
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Ge
2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	05021	Me

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4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sw

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Stored Procedure Example

The following SQL creates a stored procedure named "GetCustomersByCity" that can be used to select Customers from a particular City in the "Customers" table:

Example

Get your own SQL Server

```
DELIMITER //

CREATE PROCEDURE GetCustomersByCity
```

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```
END //

DELIMITER;
```

Here we execute the stored procedure by passing a city ('London') as a parameter, and the stored procedure returns the relevant details from the "Customers" table:

Example

```
CALL GetCustomersByCity('London');
```

Stored Procedure With Multiple Parameters

Adding multiple parameters is easy. Just list each parameter and the data type separated by a comma as shown below.

The following SQL creates a stored procedure that selects Customers from a particular City with a particular PostalCode from the "Customers" table:

Example

```
DELIMITER //

CREATE PROCEDURE GetCustomersByCity
  @City VARCHAR(50),
  @PostalCode VARCHAR(10)
BEGIN
  SELECT * FROM Customers
  WHERE City = @City AND PostalCode = @PostalCode;
END //
```

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Execute the stored procedure above as follows:

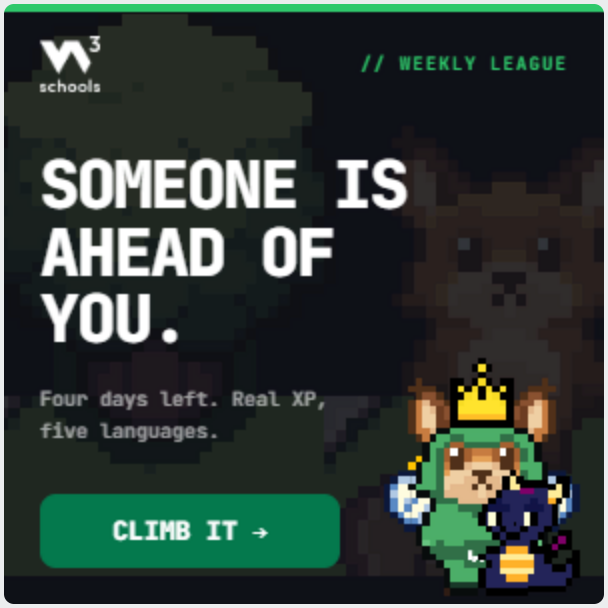
Example

```
CALL GetCustomersByCity('London', 'WA1 1DP');
```

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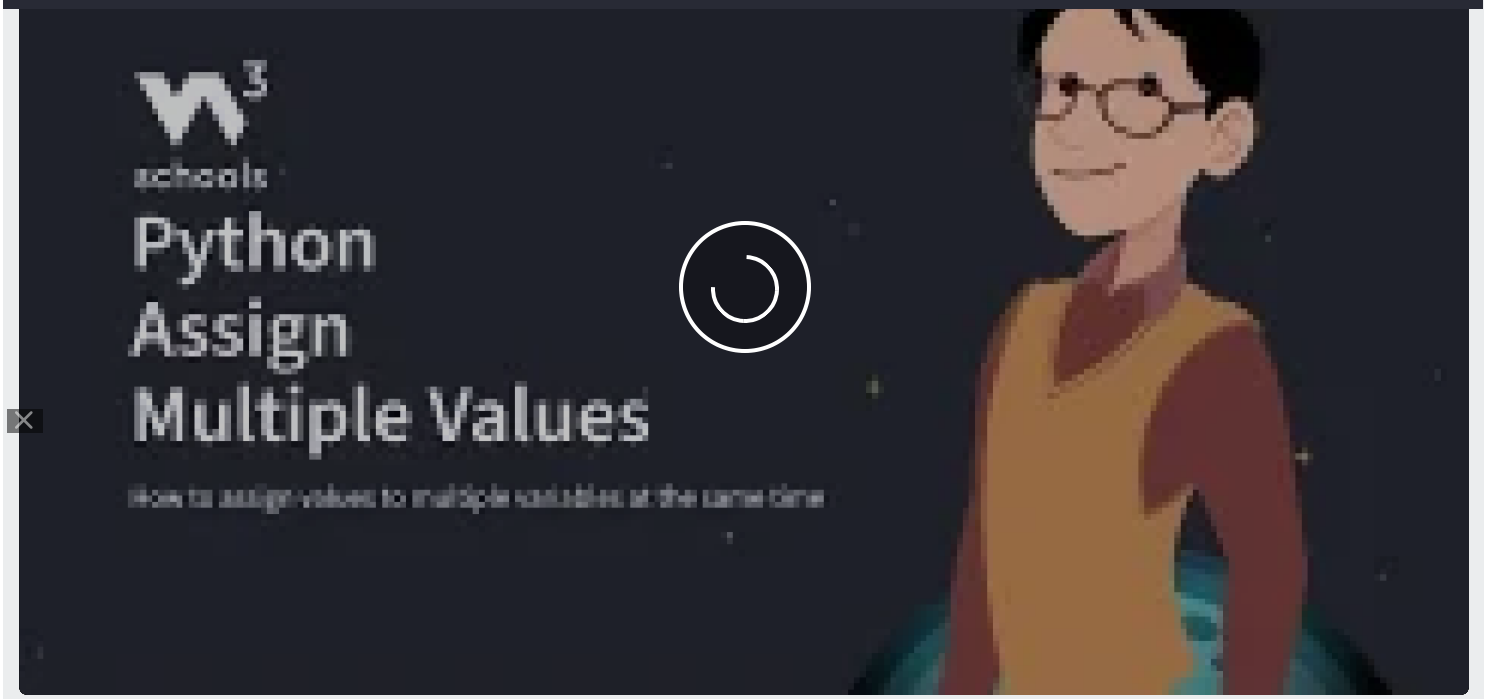
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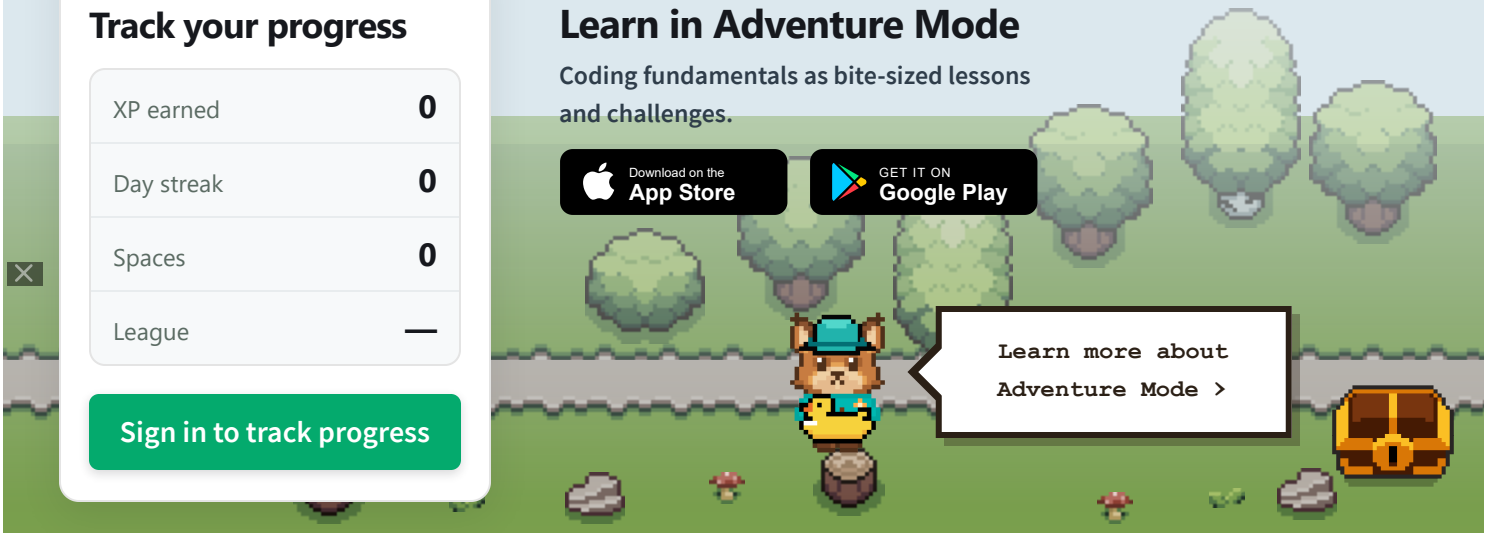
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